Activity

Objectives:

Some Groups of organisms with their distinguishing characteristics are given. Construct a cladogram, interpret and analyze the cladogram in terms of how it shows common ancestry and degrees of evolutionary relationship.

Procedure:

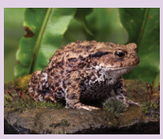
Step 1. Refer your text book and identify the characteristics of the given animals. In the data table provided, place an "x" in the box if the animal has the characteristic.

Step 2: Below the Data Table on the Worksheet, make a Venn diagram, placing animals in groups to illustrate those characteristics which different animals have in common.

Step 3: Using the Venn diagram draw a cladogram to illustrate the ancestry of these animals. The diagram should reflect the shared characteristics as time proceeds.

Step 4: Draw the Venn diagram to reflect the shared characteristics of the given animal and draw a cladogram.






Sets	Traits	Kangaroo	Lamprey	Monkey	Frog	Human	Tortoise	Fish
Set#1	Dorsal Nerve cord, Notochord							
Set#2	Paired Appendages Vertebral column							
Set#3	Paired legs							
Set#4	Amnion (Amniotic sac)							
Set#5	Mammary gland							
Set#6	Placenta							
Set#7	Canine teeth							
	Total 'X' s							

Evaluation



1. The symmetry exhibited in cnidarians is
   a. Radial
   b. Bilateral
   c. Pentamerous radial
   d. Asymmetrical
2. Sea anemone belongs to phylum
   a. Protozoa b. Porifera
   c. Coelenterata d. Echinodermata
3. The excretory cells that are found in platyhelminthes are
   a. Protonephridia **b. Flame cells**
   c. Solenocytes d. All of these
4. In which of the following organisms, self fertilization is seen.
   a. Fish b. Round worm
   c. Earthworm **d. Liver fluke**
5. Nephridia of Earthworms are performing the same functions as
   a. Gills of prawn
   b. Flame cells of Planaria
   c. Trachea of insects
   d. Nematoblasts of Hydra

XI Std Zoology Chapter-2 EM.indd 34 30/12/2022 10:32:09